

Digital Skills for the Creative Practitioner: Supporting Informal Learning of Technologies for Creativity

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ABSTRACT

The creative industries play an important role in economic, cultural and social life, and in many creative disciplines much of the workforce is made up of individual practitioners including freelancers, sole traders and small or micro enterprises. These talented creatives often need to be responsible for their own ongoing learning within challenging and ever-evolving digital and technological domains. Whether their creative practice is primarily analogue or digital, Creativity Support Tools (CSTs) and digital platforms are being adopted for use in many phases of the creative production and dissemination process. By necessity, much of the learning that creatives undertake during the adoption of technologies is self-directed, informal, and often involves peer-to-peer support. This is an important contextual factor that HCI research needs to address when developing tools and support systems for this user group. This one-day workshop will bring together participants from the HCI, creative and educational communities to discuss and share knowledge of technology learning and skills acquisition for working creatives. The workshop aims to examine ideas, strategies and experiences around supporting digital literacy, competency and confidence. The goal is to develop further collaborative research addressing support structures and frameworks in the area of informal learning about digital creativity tools for working practitioners.

CCS CONCEPTS

• **Human-centered computing**; • **Collaborative and social computing**; • **Collaborative and social computing systems and tools**; • **Interaction design** → Systems and tools for interaction design;

KEYWORDS

Creativity Support Tools, Creative Industries, Creative Practice, Digital Skills

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1 BACKGROUND

Digital skills and technical proficiencies are becoming increasingly important for practitioners in the creative industries to acquire, whether for creative production, collaboration or innovation and dissemination [1]. The pressures and changes created by the COVID-19 pandemic may have highlighted the need for digital literacy as more activities migrated to online, but this shift was part of a longstanding process that looks likely to continue and gather pace. The creative industries are known to be a particularly dynamic section of the economy, composed of a substantial number of micro-entrepreneurs and independent workers [2]. The term “CreaTech” has been coined to describe the emergence and importance of this area where digital and data skills meet creativity [3]. CreaTech encompasses digital technology for production, the creation of digital content, and the use of data-driven innovation to support creatives in their day-to-day business practice.

This digital transformation process is disruptive, and it is changing the way that designers and artists create and distribute their work, and influencing how audiences and clients access these creative products, outputs and content [4]. The consequence of this technological revolution is that acquiring and improving digital skills is becoming more necessary for those working in the creative industry sector.

Creativity Support Tools (CSTs) [5] are diverse in nature, and are aimed at varying levels of technical expertise, but even creatives working in the most analogue of media, such as crafts and textiles, are experimenting with incorporating tools such as Computer Aided Design (CAD) and Virtual and Augmented Reality (VR & AR) into

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their workflows. Visual and sound artists are learning to code in order to gain control of the production of interactive installations, and software tools like Processing [6] allow for “computer-savvy individuals” to get started with interactive and visual work. Creatives of all disciplines are using, creating and adapting online platforms for various activities, like creative livestreams which serve the purpose of teaching, making, socializing and performing [7]. New platforms and online systems are emerging to support this shift. For example, in the UK the British Broadcasting Corporation’s (BBC’s) Connected Studio (<https://www.bbc.co.uk/connectedstudio/>) network and platform offers access to tools, guidance and insight through the MakerBox tools (<https://www.bbc.co.uk/connectedstudio/how-to-get-involved>), an online community built around sharing access to technology from both inside and outside the BBC for delivering engaging broadcast experiences to reach audiences in new and diverse digital ways.

While this turn to the digital is now well established, the continuing evolution of tools and services puts pressure on to creative practitioners to keep up to speed with learning, especially for those whose practice is not always “born-digital”. This can add more pressure to already time-constrained creatives to engage with relevant digital and data services. Technology evolves very fast and often quickly becomes obsolete or discontinued by the time creatives may have mastered skills, particularly in software use. This can be an additional factor to inhibit adoption of some technologies. There is a plethora of tutorials, manuals, videos and training materials available on almost any technological topic, however, navigating and selecting appropriate and up-to-date training can become a substantial overhead for working practitioners whose time is precious. Creative practitioners perceive the navigation and appropriation of new CSTs as something guided by features, but which also includes a focus on “ease of use” in relation to the user experience and “tutorials” and “community of users” in relation to the level of support for a new tool [1].

In addition to updating competencies directly relevant to a practitioner’s own practice, there is also the need to learn digital literacy skills so that they can meaningfully communicate and collaborate with technical specialists and other creative practitioners.

There are examples of successful and well-established initiatives and movements that support learning of digital skills (in addition to the approach of trial-and-error [8]), including the open-source communities such as GitHub (<https://github.com/open-source>) that have done immense work in democratizing access to software. Fablabs (<https://www.fablabs.io/>) and makerspaces (<https://www.makerspaces.com/what-is-a-makerspace/>) have opened up access to technology tools for learning and experimenting. Software Carpentries volunteers teach “workshops around the world on the foundational skills to work effectively and reproducibly with data and code” (<https://carpentries.org/workshops/>) and academic collaborations such as FutureLearn (<https://www.futurelearn.com/>) offer accessible expert teaching online. In addition, the no-code movement offers ways to create software without having to learn to write code (<https://webflow.com/no-code>).

Alongside the growth in online offerings, in many creative disciplines the concept of the studio as a physical place for shared learning remains important [9]. It is a space that houses physical resources such as equipment, materials and software, and it is

a place where learning happens through doing and sharing [10]. A common approach used in studios is grounded in experiential learning. In the studio setting this approach is manifest in constructionism [11] which acknowledges that learning is a reconstruction of knowledge rather than a transmission; the learner constructs knowledge through objects, or ‘learning through doing’. In the studio this typically happens at both an individual and at a social level through formal and informal interactions with peers and instructors. Learning occurs through the constant iteration between concrete experiences, abstract thinking, reflective observations, and active experimentation [12].

Looking to the future, this workshop invites participants to share knowledge and experience of work related to the topics described above, and crucially, aims to speculate about the form of new tools and systems, and to shape the research agenda.

1.1 Workshop Goals

This workshop focuses on informal and collaborative digital skills provision, especially peer-to-peer learning, in the context of the creative industries. The goal of the workshop are: 1) to foreground the realities of life as an individual creative practitioner within a digital skills and creative technologies research agenda, 2) to work towards a greater understanding of how best to support professional creatives to become competent and confident digital learners, 3) to bring together researchers from academia, industry and other spheres to share, knowledge, perspectives and insights, 4) to map out an interdisciplinary research agenda, including identification of current gaps, 5) to initiate activities such as collaborative studies and publications.

The workshop will address questions including:

- **Defining and understanding skills:** What are the digital, technological and data skills that creative micro-entrepreneurs and independent practitioners will need in the future? What are the differences between digital literacy, digital competency and digital fluency in the context of creative practice? When time is limited, how much knowledge is enough? In an increasingly interdisciplinary context, how can creatives know which skills to specialise in, and which skills are needed to work collaboratively?
- **Supporting self-guided learning:** In a complex technological landscape, how can creatives navigate their own learning journeys? How can informal, peer-to-peer skills acquisition be structured? What kind of alternative support structures or frameworks would be beneficial for the individual creative in order to support communities of practice and shared learning? What are the roles of creative attributes such as curiosity in self-supported technology learning, and how can these attributes be harnessed and supported effectively?
- **Developing new tools and systems:** What kind of tools and resources should be developed to scaffold and support learning about technology? What is the potential for digital, computational tools to integrate into a primarily analogue creative workflow? What kind of tools and systems would enable creatives to support their practice to become more financially, environmentally and socially sustainable?

- **The role of physical spaces:** If there is an increasing emphasis on virtual, software-based skills, what is the role that the socio-material, studio environment plays as a resource for practitioners? Can physical, shared spaces adapt to stay relevant in an increasingly hybrid and online age?

1.2 Relevance to CHI Audience

The topics to be explored in this workshop have the potential to draw much interest from the CHI audience, including researchers interested in exploring the intersection of creativity and technology, but also extending beyond this group. Within the CHI community, there is a large body of work on CSTs, which are considered to be a grand challenge for HCI [13]. Designing such systems to support creativity at its core requires understanding the goals, motivations and realities of daily life for the intended users as well how to enable acquisition of skills necessary to use them (e.g., [14]). Moreover, the body of work on digital literacy, which has traditionally focused on teaching computing in schools, has been rapidly expanding in recent years to understand what methods can enable wider audiences to use, create with, and critically engage with technology. This is exemplified especially by recent work on understanding how to enable skills sharing, digital fabrication and digitized creative processes for wide audiences, for example in informal settings like makerspaces and FabLabs (e.g., [15]). As technology continues to permeate every aspect of life and work, understanding how to design systems, approaches and support structures to enable digital literacy for all is a timely and important endeavor. Our workshop will provide the space to explore these ideas, in the context of creative practitioners.

1.3 Participation

This workshop is aimed at a broad range of participants, including those with a research interest in; Creativity Support Tools (CSTs), digital skills acquisition, informal, collaborative or peer-to-peer learning, and professional creative practice. We wish to share a broad range of perspectives and experiences and so we welcome academics, researchers, trainers, practitioners, policy makers and others to participate.

2 WEBSITE

Prior to the workshop, the workshop website will be populated with information about the workshop, the call for participation and the submission instructions for participants. After the workshop, we will upload all participants' position papers to the website. We will also update the website with a summative account of the key outcomes of the workshop.

3 PRE-WORKSHOP PLANS

Interviews with working creatives in the UK and Denmark will be conducted, building on current work in progress. These will be shared during the workshop to feed into discussion, and to include the voices of practitioners themselves. Participants will be recruited online from the CHI and Creativity and Cognition communities, from previous workshops on Creativity Support Tools and on Digital Literacy, and from the research networks of the organizers. For example, we will share the Call for Participation and the website via

e-mail to relevant academic groups and organisations, on relevant Listservs, and on social media.

4 MODE OF DELIVERY

The intention is to hold the workshop in person only at the conference, however, the organisers understand the need to be responsive to constraints due to global events, and so the workshop could take place as a hybrid or online event if necessary.

5 WORKSHOP STRUCTURE

5.1 Schedule

Morning:

Introduction to the workshop agenda and themes by the organisers to set the context and parameters of the discussion.

Participant introductions and short presentations, giving examples of their work and ideas, thus enabling identification of common and emerging themes.

Participatory group discussion around desired goals and outcomes for the workshop, including prioritising of research questions to focus on in the afternoon sessions.

Afternoon:

This session will break up into several focused small group discussions around questions agreed upon in the morning session. Time permitting, there will be around three iterations of these discussions, enabling participants to take part in different discussions with different participants. Organisers will act as note takers and facilitators.

At the end of the small group sessions, there will be whole group sharing of main points emerging from these sessions.

Finally, there will be discussion around next steps and collaborations, for example; writing funding applications, joint research publishing, and development of best practice guidelines.

6 POST WORKSHOP PLANS

A key expected outcome of the workshop is to bring together a community of researchers interested in the intersection of digital skills and creative practice. Digital Literacy and Creativity Support Tools are two research domains that receive much interest in the CHI community however they have not to date been brought together, despite the fact that they have much potential to complement and extend each other. This workshop will formalize a community at the intersection of these two flourishing domains, and one of the post-workshop plans will be to create a means for future communication to support further collaboration and sharing of research, for example through a Slack channel. We will also plan to write a collaborative, summative position paper based on the outcomes of the workshop, outlining future directions for the questions addressed, for example in Interactions Magazine.

7 CALL FOR PARTICIPATION

This full-day, in-person workshop at CHI 2023 will focus on understanding informal and collaborative digital skills provision in the context of the creative industries. Through presentations and collaborative discussion, we will seek to foreground the realities of life as a creative practitioner within a digital skills and creative

technologies research agenda and work towards a greater understanding of how best to support professional creatives to become competent and confident digital learners.

This workshop is aimed at participants with an interest in creative practice, digital skills, and learning of technology skills through informal and peer-to-peer methods. We invite short position papers from academics, researchers, trainers, practitioners, policy makers and others. Please submit your position paper (in the SIGCHI extended abstract layout, max 4 pages, or a similar format) through the form on the workshop website. The position paper should state why you wish to participate, as well as highlight links between the workshop themes and your work or interests. Please include author names and affiliations, a short description of your research interests, and short descriptions of any examples, projects or case studies that you would like to share in the workshop. Participants will be selected based on the relevance of their position paper to the workshop themes. After the workshop the position papers will be published on the workshop website.

Please note that at least one author of each accepted paper must attend the in-person workshop at CHI 2023, and all participants must register for both the workshop and for at least one day of the conference.

8 ORGANIZERS

The organisers are actively engaged in research in this area; some have previously explored issues around creative learning spaces in the specific context of maker spaces in previous workshops [16]. Jonas Frich is an expert in CSTs, with many published works on the topic. Currently several of the UK team are involved in the UK Creative Informatics (<https://creativeinformatics.org/>) research and development programme, which aims to bring the Edinburgh city region's world-class creative industries and tech sector together, providing funding and development opportunities that enable creative individuals and organisations to explore how data can be used to drive ground-breaking new products, businesses and experiences.

Ingi Helgason (contact person) is a lecturer based at the Interaction Design Research Group, at Edinburgh Napier University. Her interests are around the experiential aspects of interacting with technology, creative and collaborative design processes, and speculative and critical design. Ingi also works on the Creative Informatics programme, with an interest in the adoption of digital technologies by creative practitioners.

Michael Smyth is an Associate Professor at the Interaction Design Research Group, Edinburgh Napier University, UK. He researches and teaches in the fields of interaction design and human computer interaction and is intrigued by the space between people and technology. Michael is a Co-Director of the Edinburgh Creative Informatics Partnership funded through the UK's Arts and Humanities Research Council Creative Industries Cluster Programme. Previously, he was the Principal Investigator on the SpeculativeEDU Project funded under the Erasmus+ initiative and the Coordinator of the FET Open UrbanIXD project. He has also worked on European Commission projects funded under Horizon2020, FP7, FP6 and FP5 initiatives. He is the co-editor of the book entitled *Digital Blur: Creative Practice at the Boundaries of Architecture, Design and Art*.

Inge Panneels is an artist and academic whose creative practice engages with space and place. She established her studio in Edinburgh in 1998 working on public art projects across the UK and has exhibited internationally. Her practice is informed by data collection, mapping, and digital making techniques developed over twelve years teaching at the Artists Designer Maker course at the University of Sunderland. This experience of the creative sector, and PhD research (AHRC) at Northumbria University into the creative mapping practices of artists, with a specific interest of how artists are charting climate change, informs her role as Research Fellow. She has published books, chapters and articles on creative mapping practices, making practices and environmental creative practices. Her current research is focussing on sustainability practices in the creative sector and the role of creativity and creative practices in engendering culture shift.

Susan Lechelt a researcher and lecturer in Human-Computer Interaction at the University of Edinburgh, interested in investigating methods for teaching novices about data and digital literacy. Her recent work has investigated the opportunities for creative practitioners to incorporate technology and data-driven innovation into their work and practice. Moreover, she has been developing methods and toolkits for conveying the creative properties of new and future technologies to creative practitioners, with the goal of inspiring more data-driven creative practice.

Jonas Frich is an Assistant Professor at Digital Design and Information Studies, Aarhus University DK. His main teaching and research topics are Creativity Support Tools and Interaction Design. He is currently working on mixed-method investigations of how practitioners leverage data in creative processes.

Eric Rawn is a PhD Student in Human-Computer Interaction at the University of California, Berkeley. He researches how practitioners (especially artists) utilize version control strategies in their process to reflect, explore, and communicate. His current interest is in building tools and theory which explore how version history can better support writing, thinking about, and learning with programs.

Bronnie McCarthy is a Producer at BBC Research & Development. Her work focuses on BBC Connected Studio MakerBox, driving the adoption of new technologies through building relationships and partnerships with both internal and external teams, offering insight and guidance on best practice for using innovative tools, creating interactive and immersive experiences, and engaging in the wider MakerBox Community through workshops, public trials, and events.

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